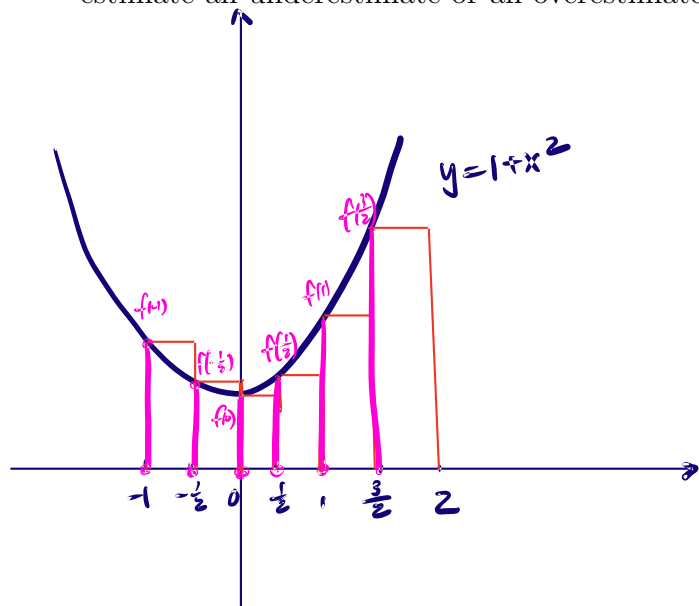


1. Estimate the area under the graph $f(x) = 1 + x^2$ from $x = -1$ to $x = 2$ using six approximating rectangles and left endpoints. Sketch the graph of the rectangles. Is your estimate an underestimate or an overestimate?



$$\Delta x = \frac{2 - (-1)}{6} = \frac{1}{2}$$

left-end points:

$$-1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}$$

Height of rectangles:

$$f(-1)=2, f(-\frac{1}{2})=\frac{5}{4}, f(0)=1, f(\frac{1}{2})=\frac{5}{4}, f(1)=2, f(\frac{3}{2})=\frac{13}{4}$$

$$L_6 = (f(-1) + f(-\frac{1}{2}) + f(0) + f(\frac{1}{2}) + f(1) + f(\frac{3}{2})) \Delta x \\ = (2 + \frac{5}{4} + 1 + \frac{5}{4} + 2 + \frac{13}{4}) \cdot \frac{1}{2} = \frac{43}{8}$$

2. Use the limit of Riemann sum definition of the definite integral to evaluate the following integral. (Note: No credit will be given if you use other method.)

1). $\int_0^2 (2 - x + x^2) dx$

$$\Delta x = \frac{2}{n}$$

Right endpoints: $x_i^* = 0 + i \cdot \Delta x = 0 + i \cdot \frac{2}{n} = \frac{2i}{n} \Rightarrow f(x_i^*) = 2 - \frac{2i}{n} + \left(\frac{2i}{n}\right)^2 = 2 - \frac{2i}{n} + \frac{4i^2}{n^2}$

$$R_n = \sum_{i=1}^n f(x_i^*) \Delta x = \sum_{i=1}^n \left(2 - \frac{2i}{n} + \frac{4i^2}{n^2} \right) \frac{2}{n} = \sum_{i=1}^n \left(\frac{4}{n} - \frac{4i}{n^2} + \frac{8i^2}{n^3} \right) = \sum_{i=1}^n \frac{4}{n} - \sum_{i=1}^n \frac{4i}{n^2} + \sum_{i=1}^n \frac{8i^2}{n^3}$$

$$= \frac{4}{n} \cdot n - \frac{4}{n^2} \left(\sum_{i=1}^n i \right) + \frac{8}{n^3} \left(\sum_{i=1}^n i^2 \right)$$

$$= 4 - \frac{4}{n^2} \cdot \frac{n(n+1)}{2} + \frac{8}{n^3} \cdot \frac{2n^3 + 3n^2 + n}{6}$$

$$= 4 - 2 \frac{n^2 + 1}{n^2} + \frac{4}{3} \frac{2n^3 + 3n^2 + n}{n^3}$$

$$= 4 - 2 \left(1 + \frac{1}{n} \right) + \frac{4}{3} \left(2 + \frac{3}{n} + \frac{1}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} R_n = 4 - 2(1 + 0) + \frac{4}{3} (2 + 0 + 0) = 4 - 2 + \frac{8}{3} = \boxed{\frac{14}{3}}$$

$$2). \int_1^2 x^3 dx$$

$\Delta x = \frac{1}{n}$, $x_i = a + i\Delta x = 1 + \frac{i}{n} = 1 + \frac{i}{n}$
 $f(x_i) = \left(1 + \frac{i}{n}\right)^3 = \dots = 1 + \frac{3i}{n} + \frac{3i^2}{n^2} + \frac{i^3}{n^3}$

$(1 + \frac{i}{n})^3 = (1 + \frac{i}{n})^2 (1 + \frac{i}{n}) = (1 + \frac{2i}{n} + \frac{i^2}{n^2}) (1 + \frac{i}{n})$
 $= 1 + \frac{2i}{n} + \frac{i^2}{n^2} + \frac{i}{n} + \frac{2i^2}{n^2} + \frac{i^3}{n^3}$
 $= 1 + \frac{3i}{n} + \frac{3i^2}{n^2} + \frac{i^3}{n^3}$

$$R_n = \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left(1 + \frac{3i}{n} + \frac{3i^2}{n^2} + \frac{i^3}{n^3}\right) \cdot \frac{1}{n} = \sum_{i=1}^n \left(\frac{1}{n} + \frac{3i}{n^2} + \frac{3i^2}{n^3} + \frac{i^3}{n^4}\right)$$

$$= \left(\sum_{i=1}^n \frac{1}{n}\right) + \frac{3}{n^2} \left(\sum_{i=1}^n i\right) + \frac{3}{n^3} \left(\sum_{i=1}^n i^2\right) + \frac{1}{n^4} \left(\sum_{i=1}^n i^3\right)$$

$$= n \cdot \frac{1}{n} + \frac{3}{n^2} \cdot \frac{n^2 + n}{2} + \frac{3}{n^3} \cdot \frac{2n^3 + 3n^2 + n}{6} + \frac{1}{n^4} \cdot \frac{n^4 + 2n^3 + n^2}{4}$$

$$= 1 + \frac{3}{2} \cdot \frac{n^2 + n}{n^2} + \frac{1}{2} \cdot \frac{2n^3 + 3n^2 + n}{n^3} + \frac{1}{4} \cdot \frac{n^4 + 2n^3 + n^2}{n^4}$$

$$= 1 + \frac{3}{2} \left(1 + \frac{1}{n}\right) + \frac{1}{2} \left(2 + \frac{3}{n} + \frac{1}{n^2}\right) + \frac{1}{4} \left(1 + \frac{2}{n} + \frac{1}{n^2}\right)$$

$$\lim_{n \rightarrow \infty} R_n = 1 + \frac{3}{2}(1+0) + \frac{1}{2}(2+0+0) + \frac{1}{4}(1+0+0) = 1 + \frac{3}{2} + 1 + \frac{1}{4} = \frac{15}{4}$$

3. Evaluate the following integral by interpreting it in terms of areas.

$$1). \int_{-2}^2 \sqrt{4-x^2} dx = 2\pi$$

$$y = \sqrt{4-x^2} \text{ with } -2 \leq x \leq 2$$

$$\Rightarrow y^2 = 4-x^2 \text{ with } y \geq 0, -2 \leq x \leq 2$$

$$\Rightarrow y^2 + x^2 = 4 \text{ with } y \geq 0, -2 \leq x \leq 2$$



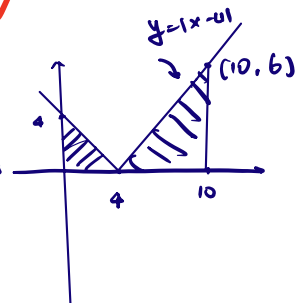
$$2). \int_0^{10} |x-4| dx$$

Shift to the right by 4 units

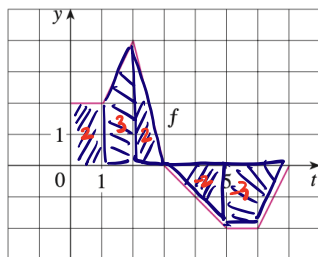
= Net area of the shaded region

$$= \frac{1}{2} \cdot 4 \cdot 4 + \frac{1}{2} \cdot 6 \cdot 6$$

$$= 8 + 18 = 26$$



4. Let $g(x) = \int_0^x f(t) dt$, where f is the function whose graph is shown. Evaluate $g(x)$ for $x = 1, 2, 3, 5$, and 7 .



$$g(1) = \int_0^1 f(t) dt = \text{net area under } f(t) \text{ on } [0, 1] = 2$$

$$g(2) = \int_0^2 f(t) dt = 2 + 3 = 5$$

$$g(3) = \int_0^3 f(t) dt = 2 + 3 + 2 = 7$$

$$g(5) = \int_0^5 f(t) dt = 2 + 3 + 2 - 2 = 5$$

$$g(7) = \int_0^7 f(t) dt = 2 + 3 + 2 - 2 - 3 = 2$$

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